

1. (10 分) 求幂级数

$$\sum_{n=1}^{\infty} \frac{x^{n^3}}{10^n}$$

的收敛域

2. (10 分) 在 $(-1, 1)$ 上展开函数

$$\arctan x + \frac{1}{2} \ln \frac{1+x}{1-x}$$

为幂级数

$$(1). \quad u_n = \frac{x^{n^3}}{10^n}$$

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{1}{10} \cdot x^{3n^2+3n+1} \right|$$

若 $-1 < x < 1$,

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = 0. \text{ 故收敛}$$

若 $|x| > 1$, $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| \neq 0$, 发散

$x=1$ 时, $\sum_{n=1}^{\infty} \frac{1}{10^n}$ 易知收敛

$x=-1$ 时, 原式 $= \sum_{n=1}^{\infty} \frac{(-1)^{n^3}}{10^n}$, $|u_n| = \frac{1}{10^n}$, 单调且 $\rightarrow 0$, 由莱布尼茨判别法知收敛

$\therefore$ 收敛域为 $[-1, 1]$

$$2. \quad (\arctan x)' = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}, \quad \therefore \arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n-1}}{2n-1}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \quad \ln(1-x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(-1)^n x^n}{n} = - \sum_{n=1}^{\infty} \frac{x^n}{n}$$

$$\therefore \ln \frac{1+x}{1-x} = \sum_{n=1}^{\infty} \left[1 + (-1)^{n-1} \right] \frac{x^n}{n} = 2 \sum_{n=1}^{\infty} \frac{x^{2n-1}}{(2n-1)}$$

$$\therefore \text{原式} = \sum_{n=1}^{\infty} \left(\frac{(-1)^{n-1}}{(2n-1)} + \frac{1}{(2n-1)} \right) x^{2n-1} = \sum_{n=1}^{\infty} \frac{2}{4n-1} x^{4n-1}, \quad -1 < x < 1$$

3. (10 分) 求瑕积分

$$\int_0^1 \sqrt{\frac{x^5}{1-x}} dx$$

的值。(本题可用 B 函数和 Γ 函数。)

4. (10 分) 判别级数

$$\int_0^1 x^{\frac{5}{2}} \cdot (1-x)^{-\frac{1}{2}} dx = B\left(\frac{7}{2}, \frac{1}{2}\right) = \frac{\Gamma(\frac{7}{2})\Gamma(\frac{1}{2})}{\Gamma(4)} = \frac{\frac{15}{8}\pi}{6} = \frac{5}{16}\pi$$

4. (10 分) 判别级数

$$\sum_{n=1}^{\infty} \frac{\sin(2n)}{n + \frac{1}{n}} \left(1 + \frac{1}{n}\right)^n$$

的敛散性。

5. (10 分) 设 E 为实数。

$$\begin{aligned} \forall n > 0, \quad \sum_{n=1}^n \sin 2n &= \frac{1}{\cos 1} \cdot \sum_{n=1}^n \sin 2n \cos 1 \\ &= \frac{1}{2\cos 1} \sum_{n=1}^n [\sin(2n+1) - \sin(2n-1)] \\ &= \frac{1}{2\cos 1} \cdot (\sin(2n+1) - \sin 1) \leq \frac{1}{\cos 1} \end{aligned}$$

$$\therefore \sum_{n=1}^{\infty} \sin 2n \text{ 有界}$$

而 $\frac{1}{n + \frac{1}{n}}$ 单减且 $\rightarrow 0$, 由 Dirichlet 判别法, $\sum_{n=1}^{\infty} \frac{\sin(2n)}{n + \frac{1}{n}}$ 收敛

而 $(1 + \frac{1}{n})^n$ 单增且 $(1 + \frac{1}{n})^n < e$

由 Abel 判别法, 级数收敛

5. (10 分) 设 E 为实数。

(a) (5 分) 求出所有的实数 E 使得

$$\int_0^{+\infty} \sum_{n=0}^{\infty} \left(\frac{(Ex)^n}{n!} e^{-x} \right) dx$$

收敛。

(b) (5 分) 求出所有的实数 E 使得

$$\sum_{n=0}^{\infty} \int_0^{+\infty} \left(\frac{(Ex)^n}{n!} e^{-x} \right) dx$$

收敛。(本小题可用 Γ 函数。)

$$(a) \quad \text{原式} = \int_0^{+\infty} e^{Ex-x} dx = \int_0^{+\infty} e^{(E-1)x} dx, \quad \therefore E-1 < 0, \quad E < 1.$$

$$(b) \quad \Gamma(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x} dx$$

$$\begin{aligned} \text{原式} &= \sum_{n=0}^{\infty} \frac{E^n}{n!} \int_0^{+\infty} x^n e^{-x} dx \\ &= \sum_{n=0}^{\infty} \frac{E^n}{n!} \Gamma(n+1) \\ &= \sum_{n=0}^{\infty} E^n \quad \therefore -1 < E < 1 \end{aligned}$$

6. (10 分) 对于每个 $x \in [0, 1]$, $n = 1, 2, \dots$, 定义

$$f_1(x) = \int_0^x \sqrt{1+t^4} dt, \quad f_{n+1}(x) = \int_0^x f_n(t) dt \quad (1)$$

证明

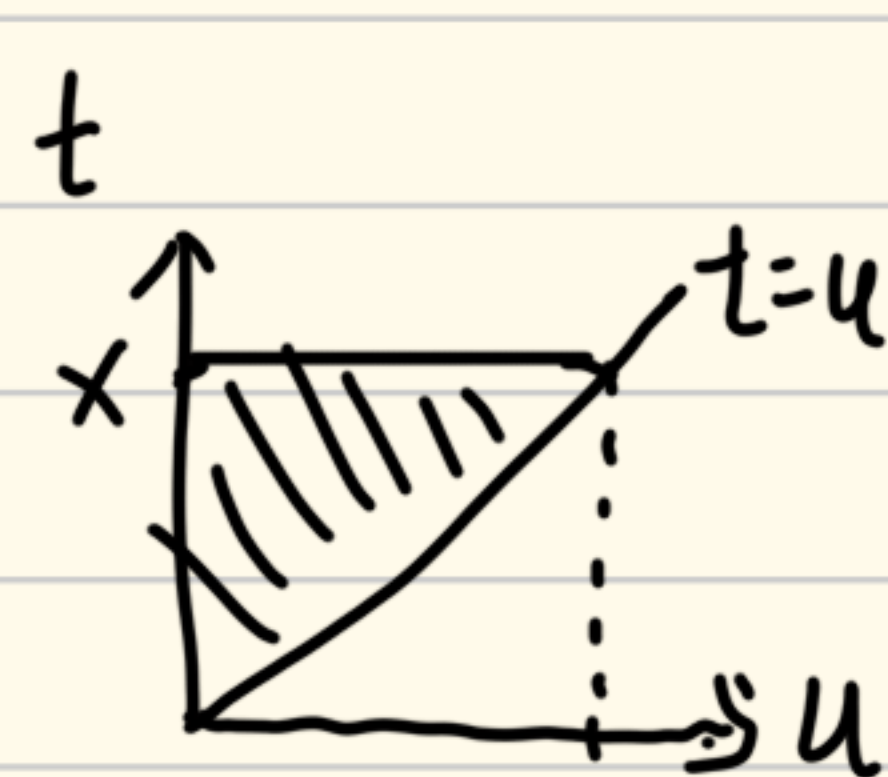
$$\sum_{n=1}^{\infty} f_n(x) \quad (2)$$

在 $[0, 1]$ 上一致收敛。

$$f_2(x) = \int_0^x f_1(t) dt = \int_0^x ds \int_0^s \sqrt{1+t^4} dt,$$

$$= \int_0^x dt \int_t^x \sqrt{1+t^4} ds$$

$$= \int_0^x (x-t) \sqrt{1+t^4} dt \quad f_2(t) = \int_0^t (t-u) \sqrt{1+u^4} du$$



$$f_3(x) = \int_0^x f_2(t) dt = \int_0^x dt \cdot \int_0^t (t-u) \sqrt{1+u^4} du$$

$$= \int_0^x du \int_u^x (t-u) \sqrt{1+u^4} dt$$

$$= \int_0^x \frac{(x-u)^2}{2} \sqrt{1+u^4} du.$$

$$\text{同理有 } f_n(x) = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} \sqrt{1+t^4} dt$$

$$\therefore \because 0 \leq t \leq x \leq 1$$

$$\therefore \sqrt{1+t^4} \leq \sqrt{2}$$

$$f_n(x) \leq \frac{\sqrt{2}}{(n+1)!} \int_0^1 (1-t)^{n+1} dt = \frac{\sqrt{2}}{n!} (1-t)^n \Big|_0^1 = \frac{\sqrt{2}}{n!}$$

而级数 $\sum_{n=1}^{\infty} \frac{\sqrt{2}}{n!}$ 对 $x \in [0, 1]$ 一致收敛

7. (15 分) 设 b 是实数。

(a) (5 分) 证明含参变量 b 的无穷积分

$$\int_0^{+\infty} e^{-x^2} x \cos(2bx) dx$$

在 $(-\infty, +\infty)$ 上一致收敛。

(b) (10 分) 证明

$$\int_0^{+\infty} e^{-x^2} \sin(2bx) dx = e^{-b^2} \int_0^b e^{-t^2} dt$$

$$(a) \quad f(b) = \int_0^{+\infty} e^{-x^2} x \cos(2bx) dx$$

$$|e^{-x^2} x \cos(2bx)| \leq e^{-x^2} x = e^{-x^2+2x-1} \frac{x}{e^{2x-1}}$$

$$\text{而 } \lim_{x \rightarrow 0} \frac{x}{e^{2x-1}} = 0. \quad \left(\frac{x}{e^{2x-1}}\right)' = \frac{e^{2x-1} - 2x \cdot e^{2x-1}}{e^{4x-2}} = \frac{(1-2x)}{e^{2x-1}},$$

$$\therefore \left| \frac{x}{e^{2x-1}} \right| \leq \frac{1}{2}$$

$$\therefore |e^{-x^2} x \cos(2bx)| \leq \frac{1}{2} e^{-(x-1)^2}$$

$$\text{而 } \int_0^{+\infty} e^{-(x-1)^2} d(x-1) = \frac{\sqrt{2}}{2},$$

故 $\forall b \in \mathbb{R}$,

$$\int_0^{+\infty} e^{-x^2} x \cos(2bx) dx \text{ 收敛}$$

(b) $f(b) = e^{-x^2} x \cos 2bx$, $f'(b) = -2e^{-x^2} x^2 \sin 2bx$, 同 (a) 知 $f'(b)$ 对 $(-\infty, +\infty)$ 一致收敛

$$\text{设 } g(b) = \int_0^{+\infty} e^{-x^2} x \cos(2bx) dx$$

$$\begin{aligned} g(b) &= \frac{1}{2b} \int_0^{+\infty} e^{-x^2} x \cdot d \sin 2bx = \frac{1}{2b} \left(e^{-x^2} x \sin 2bx \Big|_0^{+\infty} - \int_0^{+\infty} \sin 2bx (e^{-x^2} - 2x^2 e^{-x^2}) dx \right) \\ &= -\frac{1}{2b} \cdot \int_0^{+\infty} (1 - 2x^2) \cdot e^{-x^2} \sin 2bx dx \end{aligned}$$

$$\begin{aligned} g'(b) &= \int_0^{+\infty} -2e^{-x^2} x^2 \sin 2bx dx \\ &= \frac{1}{2b} \int_0^{+\infty} 2e^{-x^2} x^2 \cdot d \cos 2bx = \frac{1}{2b} \left(2e^{-x^2} x^2 \cos 2bx \Big|_0^{+\infty} - 2 \int_0^{+\infty} (x - x^3) \cdot e^{-x^2} \cos 2bx dx \right) \\ &= -\frac{1}{b} \int_0^{+\infty} (x - x^3) \cdot e^{-x^2} \cos 2bx dx \end{aligned}$$

8. (15 分)

(a) (10 分) 设 $f: \mathbb{R} \rightarrow \mathbb{R}$ 是周期为 2π 的函数, $f(x)$ 在 $(-\pi, \pi]$ 上等于 e^x 。

求出 $f(x)$ 的傅里叶级数, 并且求出 $f(x)$ 的傅里叶级数在 $x = \pi$ 处的和。

(b) (5 分) 求出级数

$$\sum_{n=1}^{\infty} \frac{1}{1+n^2} \quad (11)$$

的和。

$$8. (a) \quad a_0 = \frac{1}{2} \int_{-2}^2 e^x \cdot dx = \frac{1}{2} (e^2 - e^{-2})$$

$$a_n = \frac{1}{2} \int_{-2}^2 e^x \cos nx \, dx = \frac{1}{2n} \cdot \int_{-2}^2 e^x \sin nx \, dx = \frac{1}{n^2} \cdot \left(e^x \sin nx \Big|_{-2}^2 - \int_{-2}^2 \sin nx \cdot e^x \, dx \right)$$

$$= -\frac{1}{n^2} \cdot \int_{-2}^2 e^x \sin nx \, dx$$

$$\text{又有 } \frac{1}{2} \int_{-2}^2 e^x \cos nx \, dx = \frac{1}{2} \int_{-2}^2 \cos nx \, de^x = \frac{1}{2} \left(\cos nx \, e^x \Big|_{-2}^2 + n \int_{-2}^2 e^x \sin nx \, dx \right)$$

$$= \frac{1}{2} \left[(-1)^n (e^2 - e^{-2}) + n \int_{-2}^2 e^x \sin nx \, dx \right]$$

$$\text{联立得} \quad (n^2+1) \int_{-2}^2 e^x \cos nx \, dx = (-1)^n (e^2 - e^{-2})$$

$$\therefore \int_{-2}^2 e^x \cos nx \, dx = \frac{(-1)^n (e^2 - e^{-2})}{n^2+1}$$

$$\therefore a_n = \frac{1}{2} \cdot \frac{(-1)^n (e^2 - e^{-2})}{n^2+1}, \quad b_n = \frac{1}{2} \cdot \frac{(-1)^{n+1} n (e^2 - e^{-2})}{n^2+1}$$

$$\therefore f(x) \sim \frac{1}{2} (e^2 - e^{-2}) + \frac{1}{2} \sum_{n=1}^{\infty} \frac{(e^2 - e^{-2}) (-1)^n}{n^2+1} \cdot (\cos nx - n \sin nx), \quad -2 < x < 2$$

由 Diriklet 定理,

$$S(z) = \frac{1}{2}(e^z + e^{-z})$$

$$(b) \quad \text{令 } x=z, \quad e^z = \frac{1}{2z}(e^z - e^{-z}) + \frac{1}{z} \cdot \sum_{n=1}^{\infty} \frac{e^z - e^{-z}}{n^2 + 1}$$

$$\therefore 2z \cdot e^z = e^z - e^{-z} + 2 \cdot \sum_{n=1}^{\infty} \frac{e^z - e^{-z}}{n^2 + 1}$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n^2 + 1} = \frac{(2z-1)e^z + e^{-z}}{2 \cdot (e^z - e^{-z})}$$

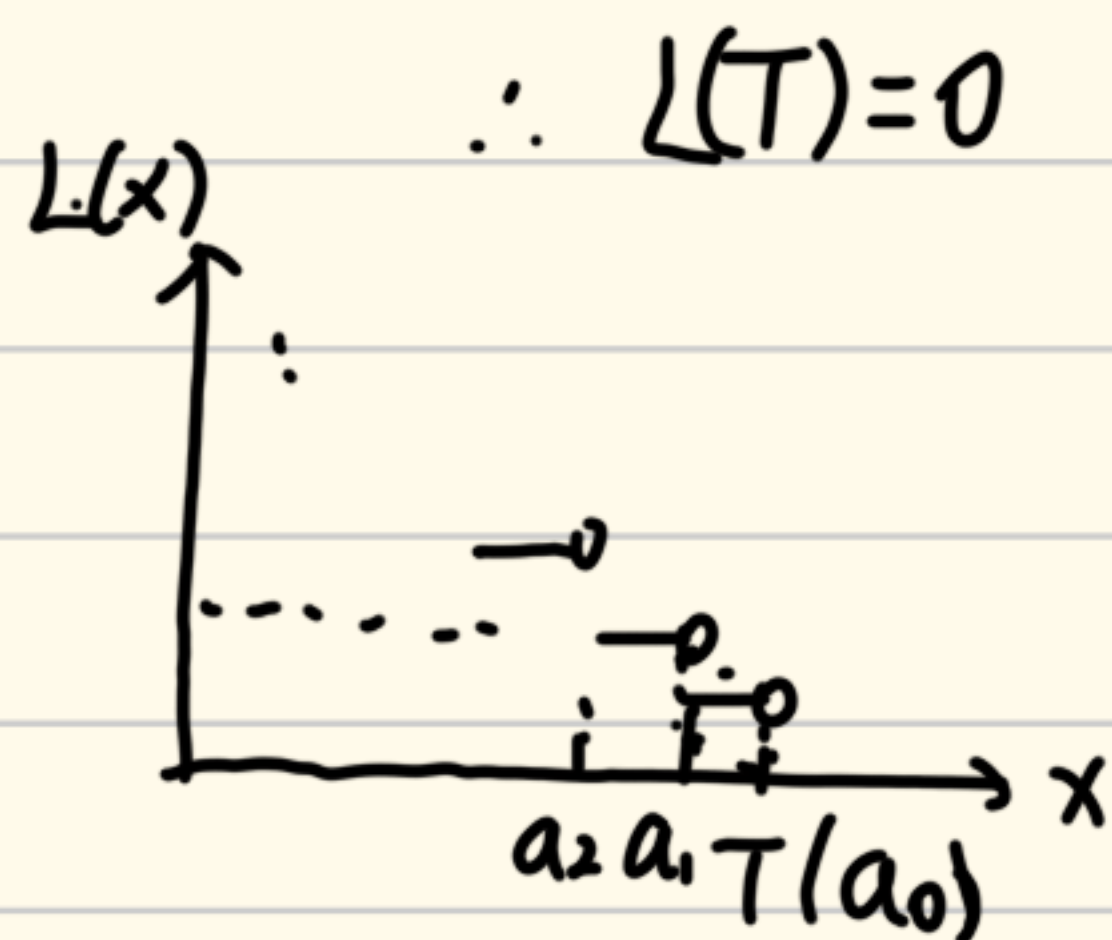
9. (10 分) 设级数 $\sum_{n=0}^{\infty} a_n$ 收敛, 每项 $a_n > 0$, T 是序列 $\{a_n\}$ 中最大项。对于每个实数 $x > 0$, 定义 $L(x)$ 是序列 $\{a_n\}$ 中大于 x 的项的个数。

(a) (2 分) 证明 0 是 $L(x)$ 的瑕点。

(b) (8 分) 证明: 瑕积分 $\int_0^T L(x)dx$ 收敛, 并且

$$\int_0^T L(x)dx = \sum_{n=0}^{\infty} a_n \quad (12)$$

$\because \sum_{n=0}^{\infty} a_n$ 收敛, $\therefore$ 可以对级数进行重排, 不妨设
 $T = a_0 > a_1 > a_2 > \dots$



$\therefore L(T) = 0$

$\because \{a_n\}$ 中存在最大项 T 且有无穷项

$\therefore \forall \varepsilon > 0, \exists N > 0$, 当 $n > N$ 时, $|a_n| < \varepsilon$

$\therefore \forall x > 0, \exists \delta > 0$, 当 $|x| < \delta$ 时, $L(x) > x$

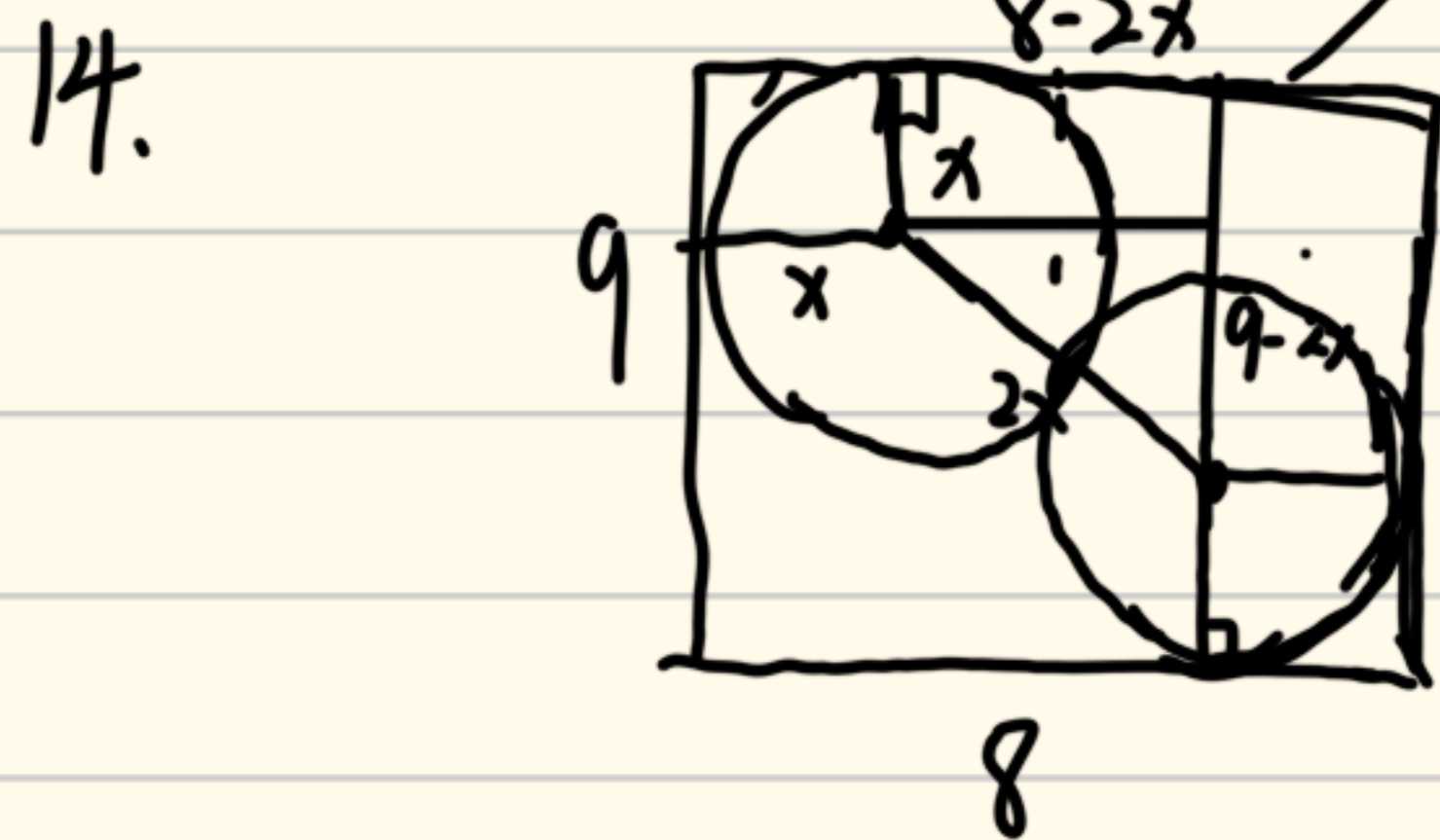
则 $\lim_{x \rightarrow 0} L(x) = +\infty$

则 0 是瑕点。

$$\begin{aligned} (b) \quad \int_0^T L(x) dx &= \sum_{n=0}^{\infty} \int_{a_n}^{a_{n+1}} L(x) dx = \sum_{n=0}^{\infty} (a_n - a_{n+1}) \cdot (n+1) \\ &= \sum_{n=0}^{\infty} a_n \end{aligned}$$

12. $\vec{a} - \vec{b} = (1, -2x)$, $\vec{a} \cdot (\vec{a} - \vec{b}) = x + 1 - 2x = 0$, $x = 1$, $\therefore |\vec{a}| = \sqrt{2}$

13. $a = 2$ $f(0) = -1 \cdot 4 = -4$



$$64 - 32x + 4x^2 + 81 - 36x + 4x^2 = 4x^2$$

$$145 - 68x + 4x^2 = 0$$

$$34 \times 5 = 170$$

$$4x^2 - 68x + 145 = 0$$

$$2$$

$$5$$

$$2$$

$$29$$

$$\therefore x = \frac{5}{2} / \frac{29}{2} (\text{舍})$$

9. $q > 0$, $S_3 = 7$, $a_3 = 1$, $S_2 = 6$, $a_1 + a_2 = 6$, $a_1 + a_1 q = 6$, $a_1(1 + q) = 6$

A $\checkmark$

$$a_1 q^2 = 1$$

B. $a_5 = a_3 \frac{1}{4} = \frac{1}{4}$ $\times$

$$\frac{1+q}{q^2} = 6, \quad 6q^2 - q - 1 = 0$$

C. S_5 是啥?

D. $S_n = \frac{4(1 - (\frac{1}{2})^n)}{\frac{1}{2}} = 8(1 - (\frac{1}{2})^n)$, $a_n = 4 \cdot (\frac{1}{2})^{n-1} = \frac{4}{2^{n-1}} = \frac{1}{2^{n-3}}$ $\checkmark$

$$q = \frac{1}{2}$$

$$a_1 = 4$$

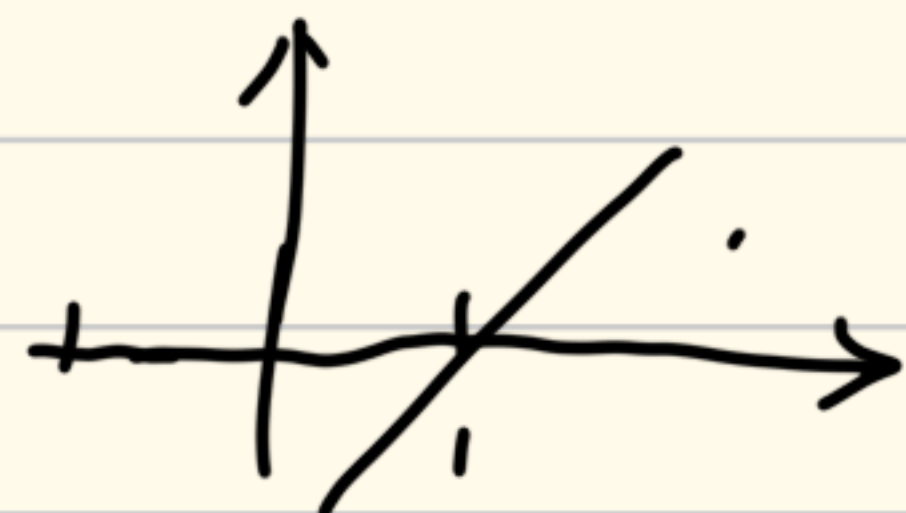
10. A. ✓

B. $f(-x) = (x^2-3)e^{-x}+2 = -f(x)$

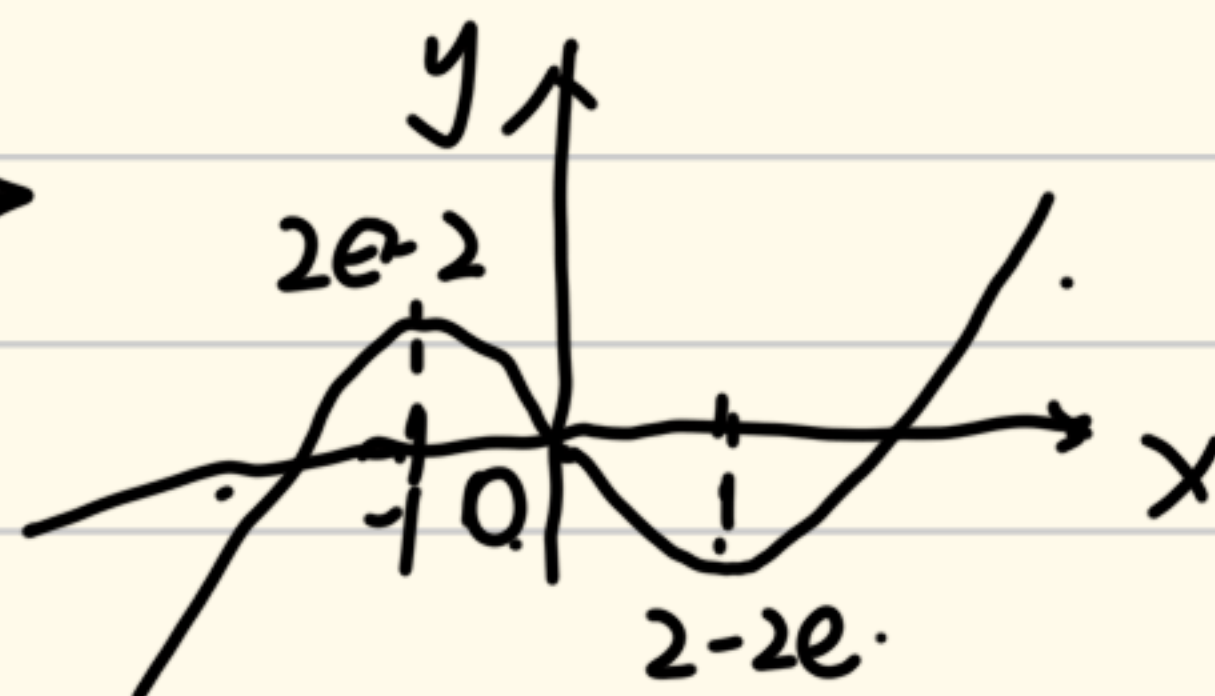
$f(x) = (3-x^2)e^{-x}-2. (x < 0)$ ✓

C. ✗ $f'(x) = e^x(2x+x^2-3) = e^x(x+3)(x-1)$

D. ✓

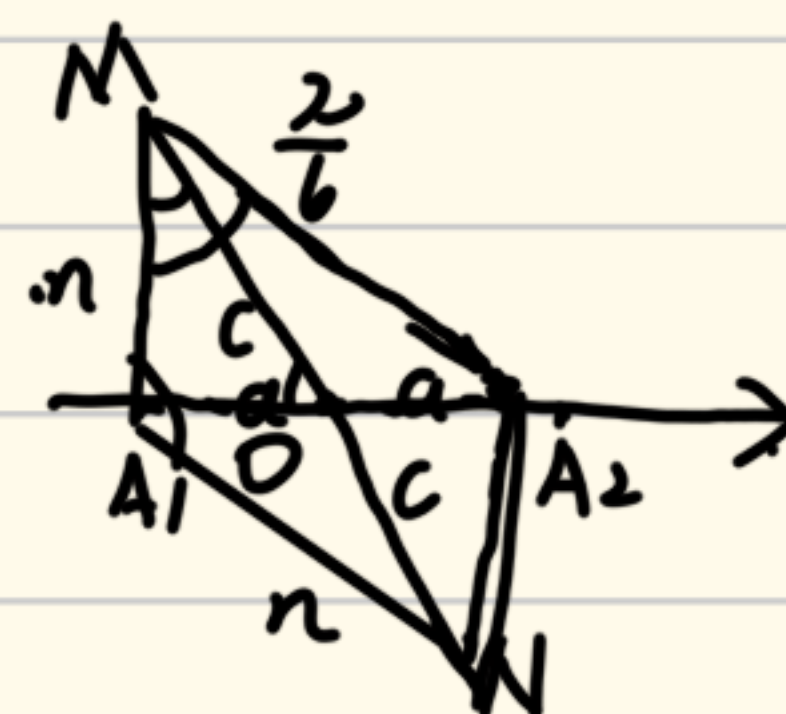
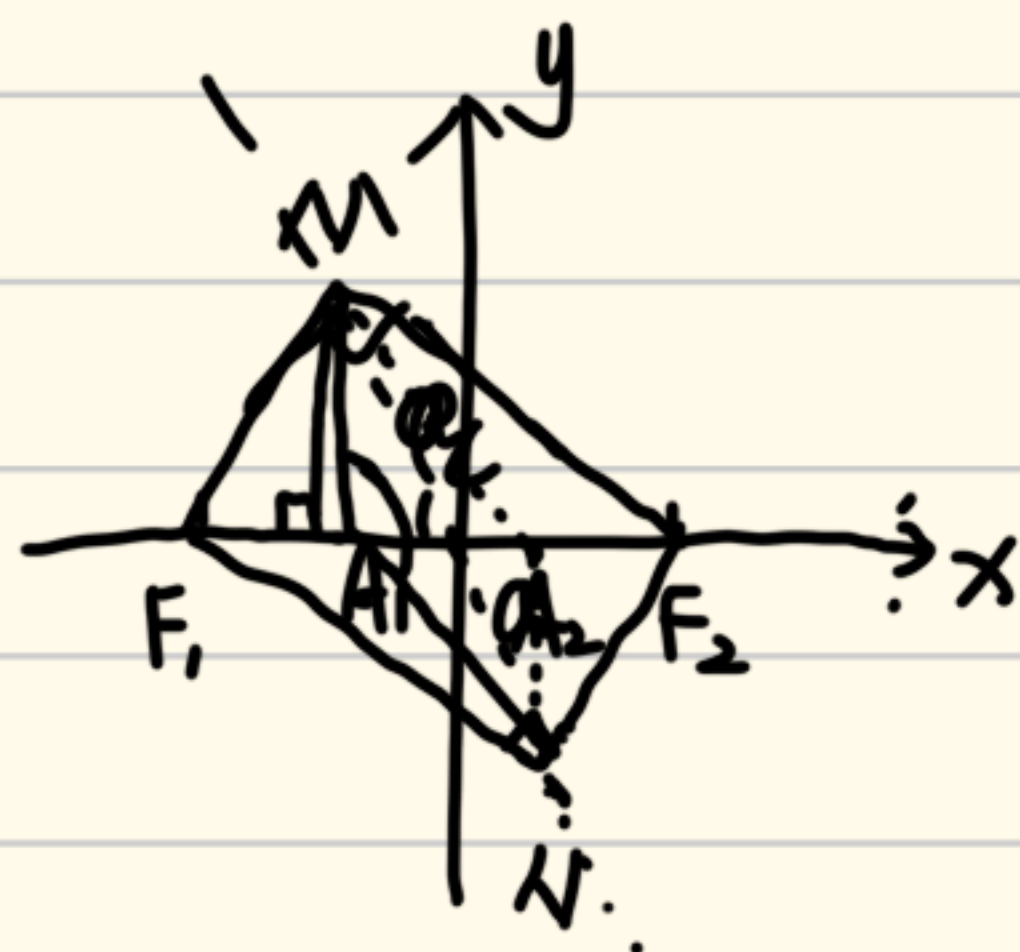


$f(1) = -2e+2$ $2e-2 > 2$



ABD

11.



A. ✓

B. ✗

C. ✓

D. ✓

$S = \frac{1}{2}mn \cdot \frac{1}{2} = \frac{1}{4}mn$

$D \leftrightarrow \therefore mn = 16\sqrt{3}$

$\tan \angle MOA_1 = \frac{1}{\frac{a^2}{b^2}+1} = \frac{b^2}{a^2+b^2} = \frac{b^2}{c^2}$

$$\begin{cases} a^2+c^2-2ac \cdot \frac{b^2}{c^2} = m^2 \\ a^2+c^2+2ac \cdot \frac{b^2}{c^2} = n^2, \quad \therefore \boxed{m^2+n^2=2a^2+2c^2} \\ m^2+n^2+\sqrt{3}mn = 4c^2 \end{cases}$$

$\therefore \sqrt{3}mn = 2c^2-2a^2 = 2b^2 \star$

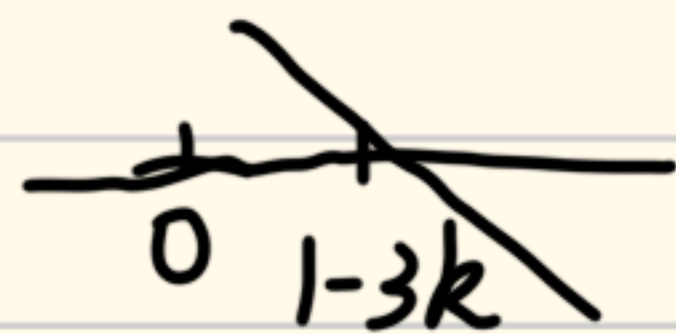
$m^2+n^2-2mn \cdot \frac{\sqrt{3}}{2} = 4a^2$

$m^2+n^2-\sqrt{3}mn = 4a^2$

即 $2b^2=48, b^2=24, b=2\sqrt{6}, a=\sqrt{2}, \therefore c=\sqrt{2}b, e=\sqrt{3}$

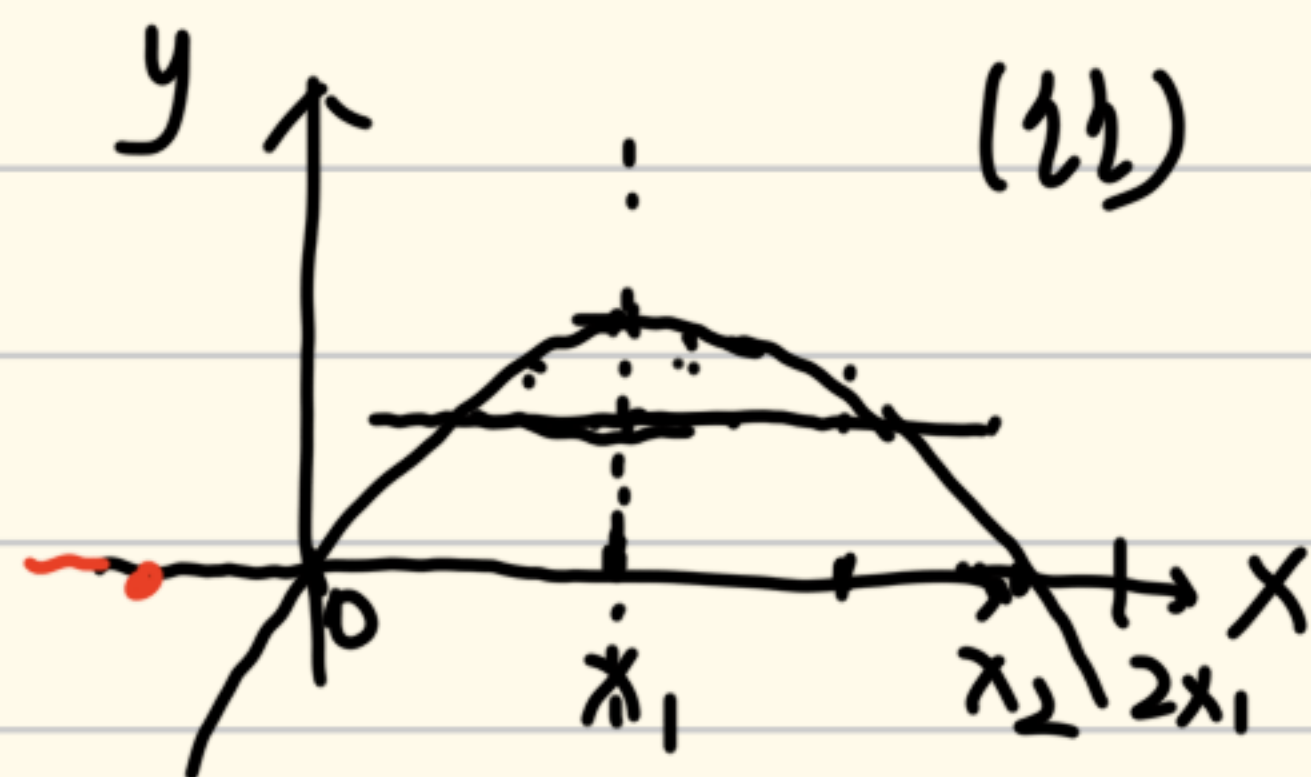
18. (1) $f'(x) = \frac{1}{x+1} - 1 + x - 3kx^2 = \frac{-x}{x+1} + x - 3kx^2 = x \cdot \left(1 - \frac{1}{x+1} - 3kx\right) = x \cdot \left(\frac{x-3kx^2-3kx}{x+1}\right)$
 $f'(0) = 0$
 $= \frac{x^2}{x+1} (1-3k-x)$

$f'(x) = 0 \Leftrightarrow x = 1-3k$, 而 $0 < 1-3k$, 故有唯一极值点 $x = 1-3k$



$\therefore f(0) = 0, \therefore f(1-3k) > 0$, 而易知 $f(9) < 0$, $\therefore$ 有唯一变点

(2) (1) $g'(t) = f'(x_1+t) + f'(x_1-t)$ $f'(x_1+t) = \frac{(x_1+t)^2}{(x_1+t+1)} (-t), f'(x_1-t) = \frac{(x_1-t)^2}{x_1-t+1} \cdot t$
 $g(0) = 0, g(x_1) = f(2x_1) < 0, f'(x_1) = 0$
 $f'(x_1-t) + f'(x_1+t) = t \cdot \left(\frac{(x_1-t)^2}{x_1-t+1} - \frac{(x_1+t)^2}{x_1+t+1} \right)$



(2) $f(2x_1) < 0$, 而 $f(x_2) = 0$

$f(x)$ 在 $(x_1, +\infty)$ $\downarrow$

$\therefore x_2 < 2x_1$

$= t \cdot \frac{(x_1-t)^2(x_1+t+1) - (x_1+t)^2(x_1-t+1)}{(x_1+1)^2 - t^2}$
 $= t \cdot \frac{(x_1^2 - 2x_1t + t^2)(x_1+t+1) - (x_1^2 + 2x_1t + t^2)(x_1-t+1)}{(x_1+1)^2 - t^2}$
 $= \frac{t}{(x_1+1)^2 - t^2} (2x_1^2t - 4x_1t + 2t^3)$
 $= \frac{t^2}{(x_1+1)^2 - t^2} (2x_1^2 - 4x_1 + 2t^2)$

$$= \frac{2-t^2}{(x_1+t)^2-t^2} \cdot (x_1-t)^2$$

19. (1) $P_3 = p^3$

$$P_4 = C_4^1 \cdot p^3 \cdot (1-p) + p^4 = 4p^3 - 4p^4 + p^4 = 4p^3 - 3p^4$$

(2) $P_4 - P_3 = 3p^3 \cdot (1-p)$

$$q_4 - q_3 = 3q^3(1-q)$$

$$\therefore \frac{p^3(1-p)}{q^3(1-q)} = \frac{p^3(1-p)}{(1-p)^3 \cdot p} = \frac{p^2}{(1-p)^2} = 4$$

$$\therefore p^2 = 4(1-p)^2, \quad p = 2(1-p)$$

$$\therefore p = \frac{2}{3}$$

(3)

